Sage Schildroth

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Education

Iowa State University, Ames, IA — Expected Graduation: May 2027
Bachelor of Science in Mechanical Engineering, Minor in Computer Science — GPA: 3.15

Experience

MACH (formerly Redshield Equipment), Ely, IA — September 2018 to Present

Mechanical Engineer (part-time), MACH — *February 2023 to Present*

- Designed an IP67 stereo camera enclosure, with vertical and horizontal mounting versions, that is now sold commercially for off-road automation.
- Build retrofit kits of sensors, hydraulics, actuators and custom components that convert off-road vehicles to full self-driving.
- Designed front and rear camera mounts, GPS mounting points and an electronics enclosure installation for an autonomous off-road vehicle.
- Fabricate metal and plastic parts using CNC machines, welders and press brakes.

Mechanical Engineer, Redshield Equipment — *September 2018 to February 2023*

- Designed and built electric unmanned vehicles for operation in dangerous environments.
- Laid out and assembled wiring harnesses for electric and autonomous vehicles.

Hearth & Home Technologies, Mt. Pleasant, IA

Mechatronics Engineering Intern — *May 2026 to August 2026*

Mechanical Engineering Intern — *June 2025 to August 2025*

- Automated a manual vacuum forming machine by integrating position sensors, pneumatic cylinders and valves, a vacuum pump, a heater and safety latches, and programming the PLC and operator interface in RSLogix 5000.
- Ran production cycles with operators and used their feedback to improve the control logic, functionality and usability of the system.
- Analyzed cycle times on a newly installed autonomous manufacturing cell and identified 1.5 seconds of potential savings in a ~25-second process (about 6%).

ATI, Millersburg, OR — June 2024 to August 2024

Mechatronics Intern

- Designed a 3-axis probe positioning system for high-temperature vacuum tensile testing that retracts the probe from the sample, preventing probe breakage when the sample fractures.
- Selected materials and components compatible with high-temperature vacuum environments.
- Collaborated with non-engineers to solve technical problems.

Research and Achievements

Wave Energy Conversion Research — January 2024 to May 2025

Undergraduate Researcher

- Created machine learning models to find more efficient ways to generate power from ocean waves.
- Tested scaled prototypes in a custom-designed wave tank.

agBOT Challenge, Rockville, IN — June 2017

Third Place, Weed & Feed Competition ($10,000 prize)

- Designed and built an autonomous robot that navigated corn rows, sprayed weeds and fertilized crops, as a solo competitor with a parent advisor.
- Competed at age 12 against university and industry teams.

Skills

CAD: SolidWorks, Autodesk Inventor | **Programming:** Python, Java, MATLAB
Controls: Studio 5000 / RSLogix 5000 (Allen-Bradley PLCs), LabVIEW
Fabrication: CNC machining, welding, press brake, wiring harnesses